

Considered there are N philosophers seated around a circular table with one chopstick between each pair of philosophers. There is one chopstick between each philosopher. A philosopher may eat if he can pick up the two chopsticks adjacent to him. One chopstick may be picked up by any one of its adjacent followers but not both. Write a program to solve the problem using process synchronization technique.

```
M /c/Users/john/OneDrive/Desktop/OSY_CD25D005
GNU nano 8.7
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

sem_t chopstick[N];

void* philosopher(void* num) {
    int id = *(int*)num;

    printf("Philosopher %d is thinking\n", id);
    sleep(1);

    // Deadlock prevention
    if (id == N - 1) {
        sem_wait(&chopstick[(id + 1) % N]);
        sem_wait(&chopstick[id]);
    } else {
        sem_wait(&chopstick[id]);
        sem_wait(&chopstick[(id + 1) % N]);
    }

    printf("Philosopher %d is eating\n", id);
    sleep(2);

    sem_post(&chopstick[id]);
    sem_post(&chopstick[(id + 1) % N]);

    printf("Philosopher %d finished eating\n", id);
    return NULL;
}

int main() {
    pthread_t phil[N];
    int ids[N];

    for (int i = 0; i < N; i++) {
        sem_init(&chopstick[i], 0, 1);
    }

    for (int i = 0; i < N; i++) {
        ids[i] = i;
        pthread_create(&phil[i], NULL, philosopher, &ids[i]);
    }

    for (int i = 0; i < N; i++) {
        pthread_join(phil[i], NULL);
    }

    return 0;
}
```

 /c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS ~

\$ cd /c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005

\$ nano phi.c

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005

\$ gcc phi.c -o phi

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005

\$./phi

Philosopher 0 is thinking

Philosopher 1 is thinking

Philosopher 2 is thinking

Philosopher 3 is thinking

Philosopher 4 is thinking

Philosopher 2 is eating

Philosopher 1 is eating

Philosopher 2 finished eating

Philosopher 3 is eating

Philosopher 3 finished eating

Philosopher 1 finished eating

Philosopher 0 is eating

Philosopher 0 finished eating

Philosopher 4 is eating

Philosopher 4 finished eating

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005

\$ |